# Development Plan — Offline Bangla Dubbing Video Player

Oct 29, 2026 · @Md. Nazmul Hossain

Version 1 is built in 6 two-week sprints, turning the spike prototype into a packaged PyQt app for Windows, macOS and Linux. This plan follows BRD v1.1, Gap Analysis v1.1, Spike v1.1 and the Rework Plan.

## Entry criteria

Development starts only when all of these are ticked:

- [ ] Spike go / no-go result is **Go** (E1 passed; no experiment hit its rework condition)
- [ ] Spike results recorded and any BRD changes approved by the product owner
- [ ] UI licence chosen: PySide6 (LGPL) or PyQt6 commercial
- [ ] Minimum and recommended hardware confirmed
- [ ] First Bangladeshi glossary drafted from spike experiment E9
- [ ] Team assigned: at least 2 developers, 1 QA, 1 product owner
- [ ] Code repository, issue tracker, CI and one GPU build machine set up

## Architecture

The app has three layers. The UI never calls a model directly: it sends jobs to the pipeline, which runs in a separate worker process so playback never freezes.

| Layer | Module | Responsibility | Requirements |
|---|---|---|---|
| UI (main process) | `player` | Video playback with mpv, audio-track and subtitle switching | BR-01, BR-11, BR-13 |
| UI | `dub_panel` | Start dubbing, progress, mode choice (pre-process / live) | BR-08, BR-09, NFR-07 |

| Layer | Module | Responsibility | Requirements |
|---|---|---|---|
| UI | `speaker_panel` | Speaker list, voice per speaker, rename / merge / split speakers | BR-05, BR-10, BR-17 |
| UI | `line_editor` | Edit a Bangla line and regenerate its voice | BR-15 |
| UI | `settings`, `model_manager` | Model download with resume and checksums, hardware detection, UI language | BR-07, BR-16, NFR-02, NFR-04 |
| Pipeline (worker process) | `separate` | Demucs voice / background split; reports quality for the fallback | BR-06 |
| Pipeline | `asr` | faster-whisper transcription and translate-to-English pivot | BR-02 |
| Pipeline | `diarize` | pyannote speaker labels, limited to the configured maximum | BR-05 |
| Pipeline | `translate` | IndicTrans2 to Bangla, then Bangladeshi glossary substitution | BR-03, BR-04a |
| Pipeline | `tts` | Indic Parler-TTS voice per speaker; swappable model for phase 2 | BR-03, BR-05, BR-04b |
| Pipeline | `timing`, `mixer` | Fit lines to slots; mix with background or lowered original audio | BR-06, NFR-05 |
| Pipeline | `scheduler` | Live mode: rolling buffer that stays ahead of playback | BR-09 |
| Storage | `cache`, `export` | Results keyed by file hash; export WAV / SRT / MKV | BR-12, BR-14 |

UI and worker talk through a job queue with progress messages. Each pipeline stage writes its result to the cache, so a crash or cancel resumes from the last finished stage.

## Tech stack and project structure

| Area | Choice |
|---|---|
| Language | Python 3.11 |
| UI | PyQt6 (or PySide6 for a closed-source LGPL app) |
| Playback | python-mpv with libmpv |
| AI | PyTorch, faster-whisper, pyannote.audio, Demucs, IndicTrans2, Indic Parler-TTS |
| Audio / video | ffmpeg |
| Worker process | `multiprocessing` with a job queue |
| Cache | SQLite index plus files per video |
| Tests | pytest, pytest-qt |
| Packaging | PyInstaller (Nuitka if startup is slow); per-OS installers |
| CI | GitHub Actions on Windows, macOS and Ubuntu runners |

```
bangla-dub/
├── app/
|   ├── main.py            # entry point
|   ├── ui/                # player, dub_panel, speaker_panel, line_editor,
settings
|   ├── pipeline/          # separate, asr, diarize, translate, tts, timing,
mixer, scheduler
|   ├── core/              # jobs, cache, export, model_manager, hardware
|   └── i18n/              # Bangla and English UI strings
├── tests/
|   ├── unit/
|   ├── integration/       # short clips through the full pipeline
|   └── fixtures/          # test clips and expected outputs
├── packaging/             # PyInstaller specs, installer scripts per OS
├── docs/
└── pyproject.toml
```

The spike's `dub_pipeline.py` is split into the `pipeline/` modules; `spike_player.py` becomes `ui/player.py`.

## Module interfaces

Every pipeline stage follows the same contract, so stages can be tested alone, cached, resumed and swapped.

```
class Stage(Protocol):
    name: str                    # "asr", "diarize", ...
    version: str                 # bump when output format or model changes

    def run(self, job: Job, inputs: StageInputs, progress: Callable[[float,
str], None]) -> StageResult:
        """Read input files, write output files into job.work_dir, return a
result object.
        Must not import UI code or open network connections."""
```

| Stage | Input | Output |
|---|---|---|
| `extract` | Video file | `original.wav` (44.1 kHz stereo) |
| `separate` | `original.wav` | `vocals.wav`, `background.wav`, separation quality score |

| Stage | Input | Output |
|---|---|---|
| `asr` | `vocals.wav` | Lines with start, end, source text, English text, detected language |
| `diarize` | `vocals.wav`, lines | Speaker label per line |
| `translate` | Lines (English) | Bangla text per line, glossary terms applied |
| `tts` | Lines, speaker → voice map | One WAV per line |
| `timing` | Lines, line WAVs | Stretch factor and start time per line |
| `mixer` | Line WAVs, background | `dubbed.wav` |
| `export` | Video, `dubbed.wav`, lines | `.bn.srt`, `.bn.mkv` |

A change in the UI (new voice, merged speaker, edited line) marks only the affected lines as dirty; `tts`, `timing` and `mixer` re-run for those lines only (BR-17, BR-15).

## Data model and cache

Each video gets a folder under the cache root, named by a SHA-256 hash of its first 64 MB plus file size, so renaming a file keeps its dub (BR-14).

```
cache/<hash>/
├── job.json         # video path, settings, stage versions, status per stage
├── lines.json       # one entry per line (below)
├── speakers.json    # speaker id → display name, voice
├── stages/          # outputs per stage
└── tts/00001.wav …  # one clip per line
```

```
{
  "id": 42, "start": 61.20, "end": 63.85,
  "source_text": "…", "english": "Where were you last night?",
  "bangla": "কাল রাতে তুমি কোথায় ছিলে?",
  "speaker": "SPK_2", "voice": "Aditi",
  "edited": false, "stretch": 1.12, "status": "ok"
}
```

A SQLite file at the cache root indexes all videos (hash, title, last opened, size) for the recent-files list and cache cleanup.

## Configuration

All model names and tuning values live in `config/default.toml`; users override them in Settings, which writes `user.toml`.

| Key | Default | Purpose |
|---|---|---|
| `asr.model` | `large-v3` (`turbo` on minimum hardware) | Whisper size |
| `diarize.max_speakers` | 6 (4 guaranteed) | BR-05 |
| `translate.model` | `ai4bharat/indictrans2-en-indic-dist-200M` | Translator |
| `translate.glossary` | `glossary/bd_bangla.tsv` | BR-04a word list |
| `tts.model` | `ai4bharat/indic-parler-tts` | Swapped in phase 2 (BR-04b) |
| `tts.voices` | Arjun, Aditi, Tapan, Rashmi, Arnav, Riya | Voice order for new speakers |
| `timing.max_speedup` | 1.35 | Sync vs naturalness |
| `separate.min_quality` | 0.6 | Below this, use the audio-lowering fallback |
| `live.buffer_seconds` | 45 | BR-09 start delay |
| `paths.cache` | OS cache folder | Where dubs are stored |

## Error handling and fallbacks

The app never stops playback because of a dubbing problem. Each fallback below is the one written into BRD v1.1.

| Situation | Behaviour | Requirement |
|---|---|---|
| One line fails in TTS or translation | That line plays original audio; marked in the line editor | NFR-06 |

| Situation | Behaviour | Requirement |
|---|---|---|
| Separation quality below threshold | Mix the dub over the original audio lowered to 25% | BR-06 |
| Bangla line too long after 1.35× speed-up | Keep it; flag it in the line editor for shortening | NFR-05, BR-15 |
| More speakers than the maximum | Merge the smallest speakers into the nearest voice; tell the user | BR-05, BR-17 |
| Hardware below recommended | Hide live mode; offer pre-process with smaller models | BR-09, NFR-02 |
| Out of GPU memory | Retry the stage on CPU or with a smaller model; show a notice | NFR-06 |
| Model file missing or corrupt | Model manager re-downloads when online; otherwise explains what is missing | BR-07, NFR-04 |
| Worker process crashes | UI restarts it and resumes from the last cached stage | — |
| Unsupported source language | Dub via English pivot and label it "best-effort" | BR-02 |

## Logging and privacy

- Logs stay on the device in the app's log folder, rotated at 10 MB, kept for 7 days.
- Logs record stage timings, errors and hardware; never transcripts, translations or file contents (NFR-03).
- No telemetry. A "Copy diagnostics" button lets users share a log voluntarily.
- The only network code is in `model_manager`; CI fails if any other module opens a connection.

## Sprint plan

Six two-week sprints (12 weeks). Start date is set once the entry criteria are met. Each sprint ends with a working build on all three platforms.

| Sprint | Goal | Scope | Requirements | Demo at sprint end |
|---|---|---|---|---|
| 1 | Foundation | Repo, CI on 3 OSes, app shell, player, worker process and job queue, config, model manager | BR-01, BR-07, NFR-04 | Play a video; download models with progress |
| 2 | Pre-process pipeline | Port spike stages into modules; cache; glossary in translate; progress in UI | BR-02, BR-03, BR-04a, BR-08, BR-14 | Dub a single-speaker English or Hindi video from the UI |
| 3 | Multi-speaker | Diarization, voice assignment, speaker panel with rename / merge / split | BR-05, BR-10, BR-17 | Dub a 4-speaker clip; merge two speakers and re-dub |
| 4 | Quality and fallbacks | Timing fit, separation fallback, line editor, audio switch, subtitles, export | BR-06, BR-11, BR-12, BR-13, BR-15, NFR-05, NFR-06 | Edit a line, regenerate, export MKV; fallback on a music-heavy clip |
| 5 | Live mode | Rolling-buffer scheduler, hardware check, auto-disable below recommended hardware | BR-09, NFR-02, NFR-10 | Start a film within 60 s on the recommended GPU |
| 6 | Release readiness | Bangla UI, installers, performance, bug fixing, user docs | BR-16, NFR-01, NFR-07, NFR-09 | Installers pass smoke tests; build handed to QA |

BR-04b (Bangladeshi accent) is phase 2: a parallel track fine-tunes Indic Parler-TTS on SUBAK.KO. The new model drops in by changing `tts.model` in the config, with no code or UI change. If live mode (BR-09) misses its target in sprint 5, it moves to version 2 without delaying release.

# Engineering practices

### Branching

- `main` is always releasable; `develop` collects finished work.
- One branch per issue: `feature/BR-05-speaker-panel`, `fix/123-sync-drift`.
- Every change goes through a pull request with 1 reviewer and green CI.

### Code standards

- Formatting and linting with ruff; type hints checked with mypy on `core/` and `pipeline/`.
- Each pipeline stage has one function: input files in, output files and a result object out. No UI imports in `pipeline/`.
- No network calls outside `model_manager`; a test fails the build if one appears.
- Model names, sizes and settings live in one config file, not in code.

### Testing during development

| Test type | What it covers | When it runs |
|---|---|---|
| Unit | Timing fit, mixer, cache, SRT writer, voice assignment | Every pull request |
| Integration | 30-second clips through the full pipeline with small models | Every pull request (CPU) |
| GPU integration | 5-minute test clips with full models; records timings | Nightly on a GPU machine |
| UI | Player controls, panels (pytest-qt) | Every pull request |
| Offline | Full run with network blocked | Nightly |

# Developer setup

1. Install Python 3.11, ffmpeg and libmpv (Ubuntu: `sudo apt install ffmpeg libmpv-dev`; macOS: `brew install ffmpeg mpv`; Windows: ffmpeg plus `libmpv-2.dll` on PATH).
2. Clone the repo and create a virtual environment: `python -m venv .venv`.
3. Install PyTorch for your hardware, then `pip install -e ".[dev]"`.

4. Accept the gated model pages on Hugging Face (pyannote segmentation 3.0 and diarization 3.1, Indic Parler-TTS, IndicTrans2), then set `HF_TOKEN`.
5. Run `python -m app.tools.fetch_models` once to download all models.
6. Start the app with `python -m app`; run tests with `pytest`.
7. For fast iteration, set `asr.model = "small"` in `user.toml` and use the 30-second clips in `tests/fixtures`.

## CI, packaging and release

| Stage | Runs on | Does |
|---|---|---|
| Lint and type check | Every pull request | ruff, mypy |
| Unit, UI, CPU integration tests | Every pull request, 3 OSes | pytest, pytest-qt, 30-second clips with small models |
| No-network check | Every pull request | Tests run with network blocked; any connection outside `model_manager` fails |
| GPU integration | Nightly, GPU machine | 5-minute fixture clips with full models; timings stored to track NFR-10 |
| Build installers | Tag on `main` | PyInstaller builds per OS |

| Platform | Installer | Notes |
|---|---|---|
| Windows | `.exe` installer (Inno Setup) | Bundles ffmpeg and libmpv; CUDA PyTorch build |
| macOS | Signed and notarised `.dmg` | Apple Silicon build; MPS PyTorch |
| Linux | AppImage, plus `.deb` | Bundles ffmpeg and libmpv |

Models are not bundled in installers (NFR-04). Versions follow semantic versioning: `1.0.0` is the first release to UAT.

## Roles and responsibilities

| Role | Responsible for |
|---|---|
| Product owner | Priorities, approves sprint demos and BRD changes, decides on fallbacks |
| Lead developer | Architecture, pipeline modules, code reviews, release builds |
| Developer | UI modules, cache, export, packaging |
| QA | Test plan, nightly GPU results, bug triage with the product owner |
| ML engineer (phase 2) | SUBAK.KO cleaning and TTS fine-tuning for BR-04b |
| UAT testers | Native Bangladeshi Bangla speakers; score sprint demos from sprint 3 |

## Development risks

| Risk | Mitigation |
|---|---|
| Library version changes break model loading | Pin versions in `pyproject.toml`; nightly GPU test catches breakage |
| Installer size too large (PyTorch, CUDA) | CPU and GPU installer variants; models downloaded separately |
| macOS notarisation of bundled libraries | Start signing in sprint 1, not sprint 6 |
| Diarization below target in real films | BR-17 tools ready in sprint 3; tune per genre |
| GPU machine is a bottleneck | Book one dedicated nightly machine before sprint 1 |

## Definition of done and handoff to QA

**A feature is done when:**

- [ ] Code is merged to `develop` after review

- [ ] Unit and integration tests pass on Windows, macOS and Linux
- [ ] It meets the requirement's acceptance condition in the BRD
- [ ] It works with the network off
- [ ] User-facing text exists in Bangla and English
- [ ] The issue is updated with what changed and how to test it

**Handoff to step 5, Testing (QA), happens when:**

- [ ] All Must requirements are done
- [ ] Installers for all three platforms are built from `main`
- [ ] Release notes list features, known issues and supported hardware
- [ ] Test clips and expected results are in `tests/fixtures`
- [ ] Open bugs: none critical, at most 5 major

## Requirement traceability

QA uses this table to build the test plan in step 5.

| Requirement | Priority | Module(s) | Sprint | Verified by |
|---|---|---|---|---|
| BR-01 Player | Must | `player` | 1 | UI tests |
| BR-02 Language detect | Must | `asr` | 2 | Integration: English, Hindi, Korean clips |
| BR-03 Speech replaced | Must | `asr`, `translate`, `tts`, `mixer` | 2 | Integration; spike E1 |
| BR-04a Bangladeshi vocabulary | Must | `translate` (glossary) | 2 | Unit (glossary); reviewer score |
| BR-04b Bangladeshi accent | Should | `tts` | Phase 2 | UAT score |
| BR-05 Voice per speaker | Must | `diarize`, `tts`, `speaker_panel` | 3 | 4-speaker fixture |
| BR-06 Keep background | Should | `separate`, `mixer` | 4 | Music fixture; fallback test |
| BR-07 Fully offline | Must | `model_manager` | 1 | No-network CI check |

| Requirement | Priority | Module(s) | Sprint | Verified by |
|---|---|---|---|---|
| BR-08 Pre-process mode | Must | All pipeline | 2 | Integration |
| BR-09 Live mode | Should | `scheduler` | 5 | Nightly GPU test |
| BR-10 Choose voices | Should | `speaker_panel` | 3 | UI tests |
| BR-11 Switch audio | Should | `player` | 4 | UI tests |
| BR-12 Export | Should | `export` | 4 | Unit; ffprobe check |
| BR-13 Subtitles | Should | `export`, `player` | 4 | Unit (SRT) |
| BR-14 Cache | Could | `cache` | 2 | Unit; replay without reprocessing |
| BR-15 Edit lines | Should | `line_editor` | 4 | UI tests |
| BR-16 Bangla UI | Could | `i18n` | 6 | Manual review |
| BR-17 Fix speakers | Should | `speaker_panel` | 3 | UI and integration |
| NFR-05 Sync | — | `timing` | 4 | Nightly: lines within 0.5 s |
| NFR-10 Speed | — | All pipeline | 5 | Nightly GPU timings |

# Change log

| Version | Date | Change |
|---|---|---|
| 1.1 | 2026-10-29 | Aligned with BRD v1.1: BR-04a/04b, BR-17, fallbacks; sprints updated |
| 1.1 | 2026-10-29 | Added module interfaces, data model, configuration, error handling, logging, developer setup, CI and packaging, roles, risks, traceability |
| 1.0 | 2026-10-29 | First version |